ALL ALUMINIUM CONDUCTORS (AAC)

AS PER IS-398-1: 1996

Code Word	Nominal Aluminium Area	Stranding and Wire Diameter	Sectional Area	Approximate Overall Diameter	Approximate Mass	Calculated Resistance at 20C, Max	Approximate Calculated Breading Load
	Sq. / mm	No. / mm	Sq. / mm	mm	Kg / km	Ω / km	KN
GNAT	25	7/2.21	26.85	6.63	74	1.096	4.52
ANT	50	7/3.10	52.83	9.3	145	0.5525	8.25
WASP	100	7/4.39	106	13.17	290	0.2752	15.96
HORNET	150	19/3.25	157.6	16.3	434	0.1825	27.7
SPIDER	240	19/3.99	237.6	19.95	654	0.1235	35.74
BUTTERFLY	300	19/4.65	322.7	23.25	888	0.09107	48.74

AS PER BRITISH SPECIFICATION BS-215-I-1970

Code Word	Conductor Size sq. mm	Standing & Wire dia no./mm	Sectional Area sq. mm	Overall Diameter mm	Approx. Weight Kg/Km	Ultimate Strength Kg	Upper Resistance at 20°c (Max.) ohm/Km
MIDGE	22	7/2.06	23.33	6.18	64	407	1.227
GNAT	25	7/2.21	26.85	6.63	74	468	1.066
ANT	50	7/3.10	52.83	9.3	145	84	0.5419
FLY	60	7/3.40	63.55	10.2	174	1010	0.4505
WASP	100	7/4.39	106	13.17	290	1632	0.2702
HORNET	150	19/3.25	157.6	16.25	434	2620	0.1825
CHAFER	200	19/3.78	213.2	18.9	587	3300	0.1349
COCKROACH	250	19/4.22	265.7	21.1	731	4120	0.1033
BUTTERFLY	300	19/4.65	322.7	23.25	888	4976	0.08916
CENTIPEDE	400	37/3.78	415.2	26.46	1145	6430	0.06944
TARANTULA	795	37/5.23	794.8	36.6	2191	12014	0.03628
BULL	865	61/4.25	864.92	38.25	2400	14169	0.03339

AS PER AMERICAN SPECIFICATION ASTM B-231 - 2016

Code Word	Conductor Size cmils/aWg	Stranding & Wire dia no./mm	Sectional Area sq. mm	Overall Diameter mm	Approx. Weight Kg/Km	Ultimate Strength Kg	Approx. Resistance at 20°c (Max.) ohm/Km
PANSY	(1)	712.78	42.39	8.32	116.8	746	0.6752
POPPY	(1/10)	7/3.12	53.48	9.4	147.4	900	0.5351
ASTER	(2/0)	7/3.50	67.42	10.51	185.8	1140	0.4245
LILY	(3)	7/2.207	26.66	6.606	73.5	515	1.076
OXLIP	(4/0)	7/4.41	107.2	13.3	295.6	1840	0.2671
TULIP	336400	1413.18	170.5	16.91	470	21290	0.169
HYACINTH	500000	3712.95	753.1	20.7	698.5	11.111	0.1138
ORCHID	636000	37/3.33	322.2	23.11	888.4	51511	0.0896
LILAC	795000	61/2.90	403.2	26.11	1112	6510	0.0719
JESSAMINE	1750000	61/4.30	886.9	38.73	2445	13500	0.0326
COWSLIP	2000000	91/3.76	1013	41.4	2793	13600	0.0284

AS PER GERMAN SPECIFICATION DIN 48 201 TEIL 5/1981

Nominal Aluminium Area sq. mm	Stranding & Wire dia no./mm	Approx. Overall Diameter mm	Sectional Area sq. mm	Approx Weight Kg/ Km	Approx. Calculated Breaking Load Kn	Calculated Resistance at 20°c (Max.) ohm/Km
16	711.70	5.10	15.89	43	2„84	I 8017
25	717.10	6.30	24.25	66	*4* 17	1 1806
35	7/2.50	7.50	34.36	94	5 78	0 8332
50	7/3 00	9.00	49.48	135	7.94	0.5786
50	19/1 80	9.00	48 35	133	8.44	0 5948
70	19/2.10	10.50	65.81	181	11.32	0.4370
95	1912.50	12.50	93.27	256	15.67	0.3084
120	19/2.80	14.00	117.00	322	18.77	0.2459
150	37/2.25	15.80	147.11	406	25:29	0.1960
185	37/2.50	17.50	181.62	500	30.54	0.1588
240	61/2.25	20.30	242.54	670	39.40	0.11920
300	61/2.50	22.50	299.43	827	47.69	0.09651
400	61/2.89	26.00	400.14	1104	60.86	0.07222
500	61/3.23	29.10	499.83	1379	74.67	0.05782
625	91/2.96	32.60	626.20	1732	95.24	0.04625
800	91/3.35	36.90	802.09	2218	118.39	0.03611
1000	91/3.74	41.10	999.71	2767	145.75	0.02897

ALL ALUMINIUM CONDUCTORS (AAC)

AS PER CANADIAN SPECIFICATION CAN/CSA - C61089 W11

Code Word	Stranding & Wire dia no./mm	Approx Overall Diameter mm	Sectional Area sq. mm	Approx. Weight Kg/Km	Approx. Calculated Breaking load Kn	Calculated Resistance at 20°c (Max.) ohm/Km
ROSE	7/1.96	5.89	21.16	58	4.07	1.351
LILY	7/2.20	6.61	26.65	73	5.05	1.072
IRIS	7/2.47	7.42	33.61	92	6.23	0.8497
PANSY	7/2.78	8.33	42.39	116	7.6	0.6739
POPPY	7/3.12	9.36	53.48	146	9.22	5.5341
ASTER	7/3.50	10.51	67.47	184	11.62	0.4236
PHLOX	7/3.93	11.8	85.03	232	14.07	0.336
OXLIP	7/4.42	13 25	107.23	293	17.75	0.2664
DALSY	7/4.96	14.9	135.16	369	22.36	0.2113
VALERIAN	19/2.91	14.57	126.71	348	22.26	0.2274
LAUREL	19/3.01	15.05	135.16	371	23.78	0.2129
PEONY	19/3.19	15.97	152	417	26.15	0.189
TULIP	19/3.38	16.11	170.45	467	29.3.8	0.1638
DAFFODIL	19/3.45	17.24	177.35	488	30.56	0.1624
CANNA	19/3.67	36.16	201.41	554	34.04	0.1427
GOLDTUFT	19/3.91	19.55	228	626	37.72	0.1263
COSMOS	19/4.02	20.12	241.68	664	40.07	0.1166
ZINNIA	19/4.12	20.6	251.35	695	41.91	0.1137
DAHLIA	19/4.35	21.73	282	774	46.69	5.1018
MEADOWSWEET	37/3.23	22.63	304	838	52.43	0.0948
ORCHID	37/3.33	23.11	372.26	688	55.57	0.0896
HEUCHERA	17/3.32	23.56	329.35	906	56.79	0.0876
VERBENA	37/3.49	24.45	354.77	978	61.16	8.0814
VIOLET	37/3.53	24:74	362.58	1000	62.34	0.0797
PETUNIA	37/3.62	48341	380	1048	64.2	0.0758
ARBIJRIS	37/3.72	26.06	402.84	1112	68.02	0.0715
ANEMONE	37/3.90	27.33	443.1	1222	73.32	0.0653
COCKSOMB	37/3.96	27.73	456.06	1257	75.48	0.0633
MAGNOLIA	37/4.06	28.55	483.42	1333	80	4.0597
HAMMED	37/4.18	29.23	506.71	1396	83.87	0.0568
BLUEBELL	37/4.24	29.72	523.68	1415	86.48	0.0551
MARIGOLD	61/3.43	30.89	563.93	1559	97.22	0.0512
HAWTHORN	61/3.55	31.95	604.26	1670	104.13	0.0479
NA.C15505	61/3.67	33.02	645.37	1781	108.79	0.045
COLUMBINE	61/3.78	34.01	684.84	1893	115.71	0.0423
CARNATION	61/389	35.03	725.1	2004	119.92	0.04
GLADIOLUS	61/4.00	35.99	765.35	2116	126.59	0.0377
COREOPSIE	61/4.10	36.91	205.68	2226	133.26	0.0358

ALUMINIUM CONDUCTORS STEEL REINFORCED (ACSR)

AS PER IS-398-II: 1996 & IS-398-V: 1992

Code Word	Nominal Aluminium Area sq. mm	Stranding & Wire Dia no./mm Alum.	Steel	Sectional Area (Aluminium) sq. mm	Total Sectional sq. mm	Approx. Overall mm	Approx. Weight Kg/Km Total Alum.	Steel		Calculated at 20°c (Max.) ohm/Km	Approx. Calculated Breaking Load Kn
MOLE	10	6/1.50	1/1.50	10.6	12.37	4.5	43	29	14	2.78	3.97
SPL. CONDUCTOR	18	6/1.96	1/1.96	18.1	21.12	5.88	73	50	23	1.618	6.74
SQUIRREL	20	6/2.11	1/2.11	20.98	24.48	6.33	85	58	27	1.394	7.61
WEASEL	30	6/2.59	1/2.59	31.61	36.88	7.77	128	87	41	0.9289	11.12
RABBIT	50	6/3.35	1/3.35	52.88	61.7	10.05	214	145	69	0.5524	18.25
RACCOON	80	6/4.09	1/4.09	78.83	91.97	12.27	319	217	102	0.3712	26.91
DOG	100	6/4.72	7/1.57	105	118 50	14.15	394	288	106	0.2792	32.41
WOLF	150	30/2.59	7/2.59	158.1	194.9	18.13	726	438	288	0.1871	67.34
PANTHER	200	30/3.00	7/3.00	212.1	261.5	21	974	587	387	0.1391	89.67
KUNDAH	400	42/3.50	7/1.96	404.1	425.2	26.88	1281	1120	161	0.07312	88.79
ZEBRA	420	54/3.18	713.18	428.9	484.5	28.62	1621	1187	434	0.06868	130.32
MOOSE	520	54/3.53	7/3.53	528.5	597	31.77	2004	1465	539	0.05595	159.6
MARKULLA	560	42/4.13	7/2.30	562.7	591.7	31.68	1781	1559	228	0.05231	120.16
BERSIMIS	690	42/4.57	7/2.54	688.9	724.4	35.04	2187	1907	280	0.04242	146.87

ALL ALUMINIUM CONDUCTORS (AAC)

AS PER IEC 61089 - 1991 (TYPE A1)

Code Number	Area	Number of Wires	Approx. Diameter		Approx. Weight	Approx. Calculated Breaking Load	Calculated Resistance at 20°c (Max.)
			Wire	Conductor			
	mm		mm	mm	kg / km	kn	ohm/km
10	10	7	1.35	4.05	27.4	1.95	2.8633
16	16	7	1.71	5.12	43.8	3.04	1.7896
25	25	7	2.13	6.40	68.4	4.50	1.1453
40	40	7	2.70	8.09	109.4	6.80	0.7158
63	63	7	3.39	10.20	172.3	10.39	0.4545
100	100	19	2.59	12.90	274.8	17.00	0.2877
125	125	19	2.89	14.50	343.6	21.25	0.2302
160	160	19	3.27	16.40	439.8	26.40	0.1798
200	200	19	3.66	18.30	549.7	32.00	0.1439
250	250	19	4.09	20.50	687.1	40.00	0.1151
315	315	37	3.29	23.00	867.9	51.97	0.0916
400	400	37	3.71	26.00	1102.0	64.00	0.0721
450	450	37	3.94	27.50	1239.8	72.00	0.0641
500	500	37	4.15	29.00	1377.6	80.00	0.0577
560	560	37	4.39	30.70	1542.9	89.60	0.0515
630	630	61	3.63	32.60	1739.3	100.80	0.0458
710	710	61	3.85	34.60	1959.1	113.60	0.0407
800	800	61	4.09	36.80	2207.4	128.00	0.0361
900	900	61	4.33	39.00	2488.3	144.00	0.0321
1000	1000	61	4.57	41.10	2759.2	160.00	0.0289
1120	1120	91	3.96	43.50	3093.5	179.20	0.0258
1250	1250	91	4.18	46.00	3452.6	200.00	0.0231
1400	1400	91	4.43	48.70	3866.9	224.00	0.0207
1500	1500	91	4.58	50.40	4143.1	240.00	0.0193

ALUMINIUM ALLOY CONDUCTORS STEEL REINFORCED (AACSR)

AS PER IEC 61089 - 1991

Code Number	Steel Ratio	Areas			No. of Wires		Wire Diameter		Diameter		Linear Mass	Rated Strength	D.C. Resistance
		Alum.	Steel	Total	Al	Steel	Al	Steel	Core	Cond.			
	%	mm^2	mm^2	mm^2			mm	mm	mm	Mm	kg/km	kn	ohm/km
16	17	18.4	3.07	21.5	6	1	1.98	1.98	1.98	5.93	74.4	9.02	1.7934
25	17	28.8	4.80	33.6	6	1	2.47	2.47	2.47	7.41	116.2	13.96	1.1478
40	17	46.0	7.67	53.7	6	1	3.13	3.13	3.13	9.38	185.9	22.02	0.7174
63	17	72.5	12.1	84.6	6	1	3.92	3.92	3.92	11.8	292.8	34.68	0.4555
100	6	115	6.39	121	18	1	2.85	2.85	2.85	14.3	366.4	41.24	0.2880
125	6	144	7.99	152	18	1	3.19	3.19	3.19	16.0	458.0	51.23	0.2304
125	16	144	23.4	167	26	7	2.65	2.06	6.19	16.8	579.9	69.86	0.2310
160	6	184	10.2	194	18	1	3.61	3.61	3.61	18.0	586.2	65.58	0.1800
160	16	184	30.0	214	26	7	3.00	2.34	7.01	19.0	742.3	88.52	0.1805
200	6	230	12.8	243	18	1	4.04	4.04	4.04	20.2	732.8	81.97	0.1440
200	16	230	37.5	268	26	7	3.36	2.61	7.83	21.3	927.9	110.64	0.1444
250	10	288	28.3	316	22	1	4.08	2.27	6.80	23.1	1013.5	117.09	0.1154
250	16	288	46.9	335	26	7	3.75	2.92	8.76	23.8	1159.8	138.31	0.1155
315	7	363	25.1	388	45	7	3.20	2.14	6.41	25.6	1196.5	136.28	0.0917
315	16	363	59.0	422	26	7	4.21	3.28	9.83	26.7	1461.4	171.90	0.0917
400	7	460	31.8	492	45	7	3.61	2.41	7.22	28.9	1519.4	172.10	0.0722
400	13	460	59.7	520	54	7	3.29	3.29	9.88	29.7	1738.3	201.46	0.0723
450	7	518	35.8	554	45	7	3.83	2.55	7.66	30.6	1709.3	193.61	0.0642
450	13	518	67.1	585	54	7	3.49	3.49	10.5	31.5	1955.6	226.64	0.0643
500	7	575	39.8	615	45	7	4.04	2.69	8.07	32.3	1899.3	215.12	0.0578
500	13	575	74.6	650	54	7	3.68	3.68	11.1	33.2	2172.9	251.82	0.0578
560	7	645	44.6	689	45	7	4.27	2.85	8.54	34.2	2127.2	240.93	0.0516
560	13	645	81.6	726	54	19	3.90	2.34	11.7	35.1	2420.9	283.21	0.0516
630	4	725	31.3	756	72	7	3.58	2.39	7.16	35.8	2248.0	249.62	0.0459
630	13	725	91.8	817	54	19	4.13	2.48	12.4	37.2	2723.5	318.61	0.0459
710	4	817	35.3	852	72	7	3.80	2.53	7.60	38.0	2533.4	281.32	0.0407
710	13	817	104	921	54	19	4.39	2.63	13.2	39.5	3069.4	359.06	0.0407
800	4	921	39.8	961	72	7	4.04	2.69	8.07	40.4	2854.6	316.98	0.0361
800	8	921	76.7	997	84	7	3.74	3.74	11.2	41.1	3145.1	356.03	0.0362
900	4	1036	44.8	1081	72	7	4.28	2.85	8.56	42.8	3211.4	356.60	0.0321
900	8	1036	86.3	1122	84	7	3.96	3.96	11.9	43.6	3538.3	400.53	0.0322
1000	8	1151	93.7	1245	84	19	4.18	2.51	12.5	45.9	3916.8	446.37	0.0289
1120	8	1289	105	1394	84	19	4.42	2.65	13.3	48.6	4386.8	499.93	0.0258